chapter 4 Python Type Conversion – Overview

Type	Draft
Status	Not started
Project	My Python Journey

types of conversions in python overview

Definition: Type conversion is the process of converting one data type into another data type.

Python has two main types of type conversion.

1. Implicit Type Conversion

Python automatically converts one data type into another without the programmer explicitly converting it.

Example:

```
a = 10b = 2.5result = a + bprint(result)print(type(result))
```

Output:

```
12.5
<class 'float'>
```

Here, Python automatically converts `int` into `float`.

2. Explicit Type Conversion (Type Casting)

The programmer manually converts one data type into another using built-in functions.

Example:

```
a = "100"b = int(a)print(b)print(type(b))
```

Output:

```
100
<class 'int'>
```

Here, the string `"100"` is converted into an integer using `int()`.

Important Type Conversion Functions

Function	Purpose	Example
`int()`	Converts into integer	`int("10")`
`float()`	Converts into float	`float("10.5")`
`str()`	Converts into string	`str(100)`
`bool()`	Converts into Boolean	`bool(1)`
`list()`	Converts into list	`list((1, 2))`
`tuple()`	Converts into tuple	`tuple([1, 2])`
`set()`	Converts into set	`set([1, 2, 2])`
`dict()`	Creates a dictionary from key-value pairs	`dict([("a", 1)])`

Difference between Implicit and Explicit

Implicit	Explicit
Automatic conversion	Manual conversion
Done by Python	Done by programmer
No conversion function required	Uses conversion functions
Example: `10 + 2.5`	Example: `int("10")`

Important points to remember

1. Implicit conversion is performed automatically.
2. Explicit conversion is performed manually.
3. Not every conversion is possible.
4. Invalid conversions can produce errors such as `ValueError` or `TypeError`.

Important interview questions

1. What is type conversion in Python?
2. How many types of type conversion are there?
3. What is implicit type conversion?
4. What is explicit type conversion?
5. What is the difference between implicit and explicit conversion?
6. What is the use of `int()`, `float()` and `str()`?
7. Can we convert a string into an integer?
8. What happens if we write `int("Hello")`?